

Physics

1. The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 the ($g = 10 \text{ ms}^{-2}$):

A. $\theta_0 = \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

B. $\theta_0 = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$

C. $\theta_0 = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

D. $\theta_0 = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$

Ans: D

Given, $g = 10 \text{ m/s}^2$

As we know that equation of trajectory

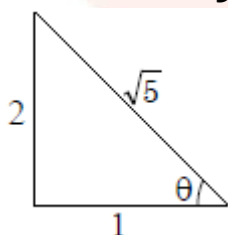
$$y = x \tan \theta \left(1 - \frac{x}{R}\right) \dots (1)$$

Given equation of trajectory: $y = 2x - 9x^2 = 2x$

$$\left(1 - \frac{9x}{2}\right) = 2x \left(1 - \frac{x}{\frac{2}{9}}\right) \dots (2)$$

Comparing equation (1) & (2)

$$\tan \theta = 2 \text{ \& } R = \frac{2}{9}$$



$$\cos \theta = \frac{1}{\sqrt{5}} \Rightarrow \theta = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$$

$$\Rightarrow \frac{u^2 \sin 2\theta}{g} = \frac{2}{9}$$

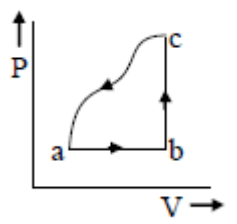
$$\Rightarrow \frac{u^2 \left[\frac{2 \tan \theta}{1 + \tan^2 \theta} \right]}{10} = \frac{2}{9}$$

$$\Rightarrow u^2 \left[\frac{4}{5} \right] = \frac{2}{9} \times 10$$

$$\Rightarrow u^2 = \frac{2 \times 10 \times 5}{9 \times 4}$$

$$\Rightarrow u = \frac{10}{6} = \frac{5}{3} \text{ m/s}$$

2. A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is -180 J . The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is.



A. 120 J

B. 130 J

C. 140 J

D. 100 J

Ans: B

For the process $(c \rightarrow a)$

The change in the internal energy of the gas, $\Delta U_{ca} = -180 \text{ J}$

$\Delta U_{ac} = +180 \text{ J}$ (state function)

As we know that

$\therefore$ The process $(a \rightarrow b)$ $\Delta Q_{a \rightarrow b} = 250$

for process $(b \rightarrow c)$ $\Delta Q_{b \rightarrow c} = 60$

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q_{a \rightarrow c} = \Delta U_{a \rightarrow c} + \Delta W_{a \rightarrow c}$$

$$\Delta Q_{a \rightarrow b} + \Delta Q_{b \rightarrow c} = 180 + \Delta W_{a \rightarrow c}$$

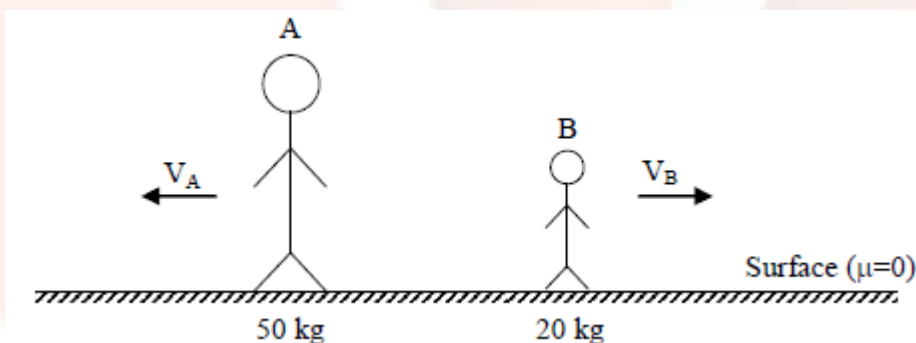
$$250 + 60 = 180 + \Delta W_{a \rightarrow c}$$

$$\Delta W_{a \rightarrow c} = 310 - 180 = 130 \text{ J}$$

3. A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at a speed of 0.70 ms^{-1} with respect to the man. The speed of the man with respect to the surface is:

A. 0.47 ms^{-1} B. 0.28 ms^{-1}
C. 0.20 ms^{-1} D. 0.14 ms^{-1}

Ans: C



Given that, mass of man = 50 kg

mass of son = 20 kg

$$\vec{V}_{B/A} = 0.7 \text{ m/s}$$

Relative speed of son with respect to man

$$\Rightarrow \vec{V}_{B/S} - \vec{V}_{A/S} = 0.7$$

$$\Rightarrow V_B - (-V_A) = 0.7$$

$$\Rightarrow V_B + V_A = 0.7$$

$$\Rightarrow V_A + V_B = 0.7$$

By using conservation of momentum

$$P_C = P_f$$

$$\Rightarrow 0 = 20(+V_B) + 50(-V_A)$$

$$\Rightarrow 2V_B = 5V_A$$

$$\Rightarrow V_B = \frac{5V_A}{2}$$

$$\Rightarrow V_A + \frac{5V_A}{2} = 0.7$$

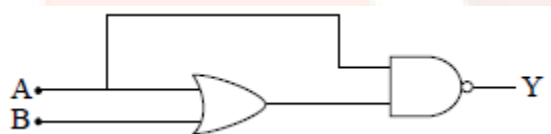
$$\Rightarrow \frac{7V_A}{2} = 0.7$$

$$\Rightarrow V_A = 0.7 \left(\frac{2}{7} \right)$$

The speed of man with respect to surface

$$\Rightarrow V_A = 0.2 \text{ m/s}$$

4. The truth table for the circuit given in the fig. is:



A.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

B.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

C.

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

D.

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

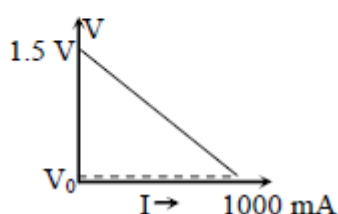
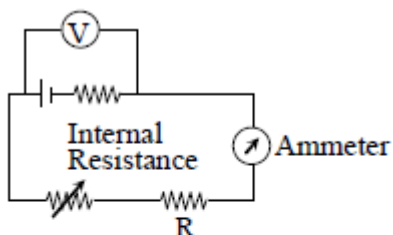
Ans: A

$$Y = \overline{A} \cdot (A + B) = \overline{A} + \overline{A} \cdot B$$

$$\Rightarrow Y = \overline{A} + \overline{A} \cdot B = \overline{A}(1 + B)$$

$$\Rightarrow Y = \overline{A}$$

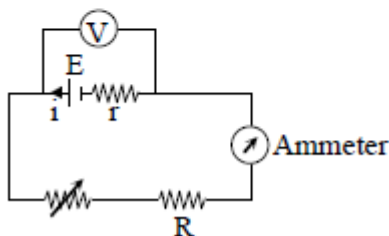
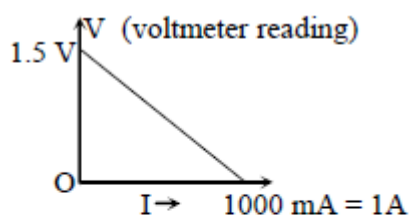
5. To verify Ohm's law, a student connects the voltmeter across the battery, as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained:



If V_0 is almost zero, identify the correct statement:

- A. The emf of the battery is 1.5 V and its internal resistance is 1.5Ω .
- B. The potential difference across the battery is 1.5 V when it sends a current of 1000 mA.
- C. The emf of the battery is 1.5 V and the value of R is 1.5Ω .
- D. The value of the resistance R is 1.5Ω .

Ans: C



Given, circuit in a series combination of internal resistance of cell (r) and external resistance R.

∴ Effective resistance in the circuit $R_{\text{eff}} = r + R$

When voltmeter reading is zero

$$\Rightarrow E - ir = 0$$

$$\Rightarrow E - \left(\frac{E}{R+r} \right) r = 0$$

$$\Rightarrow 1 - \frac{r}{R+r} = 0$$

$$\Rightarrow R + r - r = 0$$

$$\Rightarrow R = 0 : \text{when voltmeter reading is zero}$$

$$\Rightarrow R_{\text{eq}} = r \text{ (for circuit)}$$

$$i = \frac{E}{r}$$

$$\Rightarrow 1000 \text{ mA} = \frac{1.5}{r} \quad (E = 1.5 \text{ from graph})$$

$$\Rightarrow 1 = \frac{1.5}{r}$$

$$\Rightarrow r = 1.5 \Omega$$

6. A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product $t_1 t_2$ is :

A. $2R/g$

B. R/g

C. $R/2g$

D. $R/4g$

Ans: A

For θ & $(90 - \theta)$ angle of projection, horizontal range will be same

Time of flight for $\theta : t_1 = \frac{2u \sin \theta}{g}$

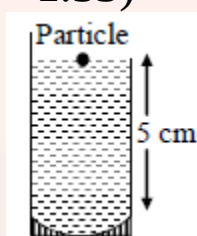
Time of flight for $90 - \theta : t_2 = \frac{2u \sin(90 - \theta)}{g} = \frac{2u \cos \theta}{g}$

$$\Rightarrow t_1 t_2 = \frac{4u^2 \sin \theta \cos \theta}{g^2} = \frac{2u^2}{g} \left(\frac{\sin 2\theta}{g} \right) = \frac{2}{g} \left(\frac{u^2 \sin 2\theta}{g} \right) = \frac{2R}{g}$$

So, the product $t_1 t_2 = \frac{2R}{g}$

7. A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to:

(Refractive index of water = 1.33)



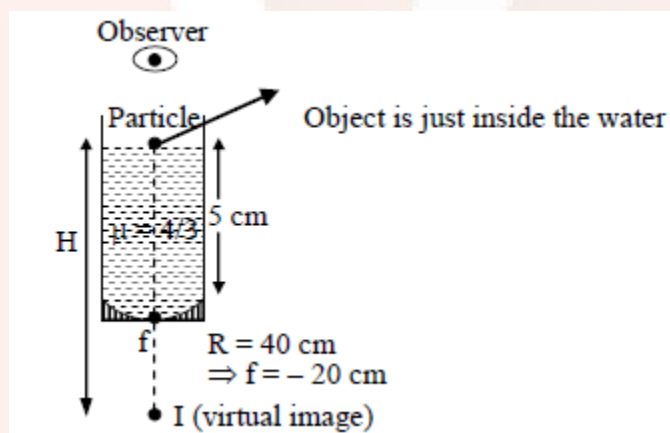
A. 6.7 cm

B. 8.8 cm

C. 11.7 cm

D. 13.4 cm

Ans: B



In the given situation

$$\text{focal length, } f = \frac{-R}{2} = \frac{-40}{2} = -20 \text{ cm}$$

Now, using mirror formula

$$\frac{1}{V} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{V} = \frac{1}{5} - \frac{1}{20}$$

$$\Rightarrow \frac{1}{V} = \frac{4-1}{20} = \frac{3}{20}$$

$$\Rightarrow V = \frac{20}{3} \text{ cm}$$

So, distance of object

$$H = 5 + \frac{20}{3} = \frac{35}{3} \text{ cm}$$

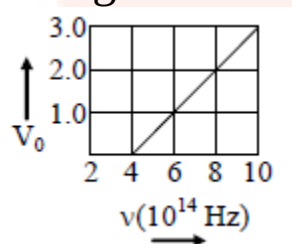
$$H_{\text{apparent}} = \frac{H}{\mu} = \frac{\left(\frac{35}{3}\right)}{\left(\frac{4}{3}\right)} = \frac{35}{3} \times \frac{3}{4} = \frac{35}{4} = 8.8 \text{ cm}$$

8. The stopping potential V_0 (in volt) as a function of frequency (ν) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be:

(Given: Plank's constant

$h = 6.63 \times 10^{-34} \text{ Js}$, electron

Charge $e = 1.6 \times 10^{-19} \text{ C}$)



A. 1.95 eV

B. 1.66 eV

C. 2.12 eV

D. 1.82 eV

Ans: B

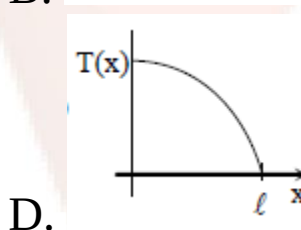
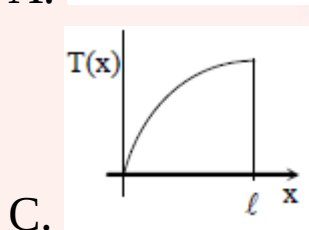
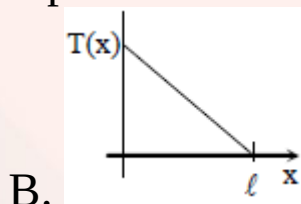
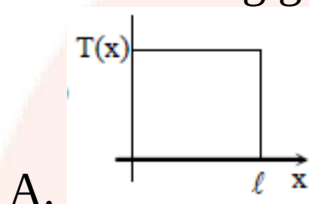
Given, Planck's constant $h = 6.63 \times 10^{-34} \text{ J-s}$

$e = 1.6 \times 10^{-19} \text{ C}$

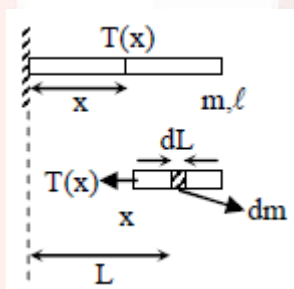
From graph, threshold frequency, $V_0 = 4 \times 10^{14} \text{ Hz}$ (where $V_0 = 0$)

$$\begin{aligned}\text{Work function} &= \phi_0 = h\nu_0 \\ &= \frac{(6.63 \times 10^{-34})(4 \times 10^{14})}{1.6 \times 10^{-19}} \text{ eV} = 1.66 \text{ eV}\end{aligned}$$

9. A uniform rod of length l is being rotated in a horizontal plane with a constant angular speed about an axis passing through one of its ends. If the tension generated in the rod due to rotation is $T(x)$ at a distance x from the axis, then which of the following graphs depicts it most closely ?



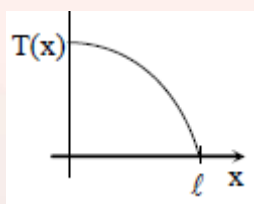
Ans: D



To find tension at x distance from fixed end, let us assume on element of the dL length and dm mass.

$$\left(\because dm = \frac{m}{l} dL \right)$$

$$\begin{aligned}
 T(x) &= \int_x^\ell (dm) \omega^2 L \\
 &= \int_x^\ell \left[\frac{m}{\ell} dL \right] \omega^2 L \\
 &= \frac{m}{\ell} \omega^2 \left[\frac{L^2}{2} \right]_x^\ell \\
 T(x) &= \frac{m\omega^2}{2\ell} (\ell^2 - x^2) \\
 T &\propto -x^2
 \end{aligned}$$



10. Which of the following combinations has the dimension of electrical resistance (ϵ_0 is the permittivity of vacuum and μ_0 is the permeability of vacuum)?

A. $\sqrt{\frac{\mu_0}{\epsilon_0}}$

B. $\frac{\mu_0}{\epsilon_0}$

C. $\frac{\epsilon_0}{\mu_0}$

D. $\sqrt{\frac{\epsilon_0}{\mu_0}}$

Ans: A

As we know that, $t = RC \dots(1)$

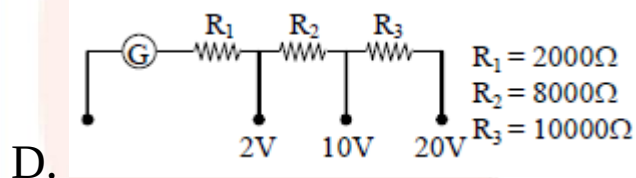
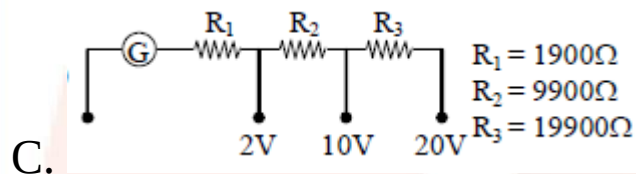
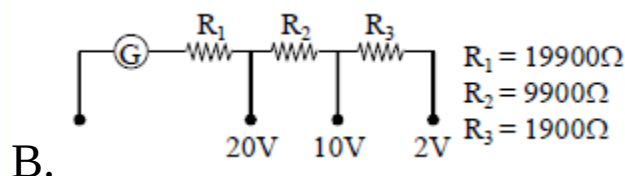
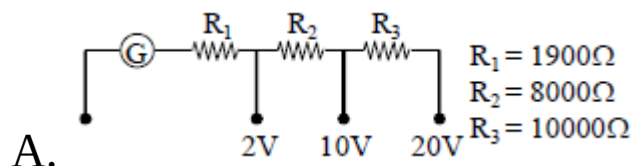
$$t = \frac{L}{R} \dots(2)$$

from (1) and (2)

$$R^2 \frac{C}{L} = 1 \Rightarrow R = \sqrt{\frac{L}{C}}$$

$$R = \sqrt{\frac{\mu_0}{\epsilon_0}} = (ML^2T^{-3}A^{-2})$$

11. A galvanometer of resistance $100\ \Omega$ has 50 divisions on its scale and has sensitivity of $20\ \mu\text{A}/\text{division}$. It is to be converted to a voltmeter with three ranges, of $0\text{--}2\text{V}$, $0\text{--}10\text{V}$ and $0\text{--}20\text{V}$. The appropriate circuit to do so is:



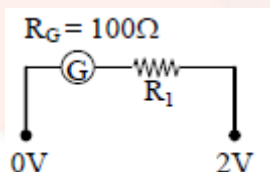
Ans: A

Given,

Sensitivity of galvanometer = $20\ \mu\text{A}/\text{div}$

Total divisions in scale of galvanometer = 50

$$\Rightarrow \text{maximum current through galvanometer} = I_{\text{max}} = (50)(20\ \mu\text{A}) = 10^{-3}\ \text{A}$$



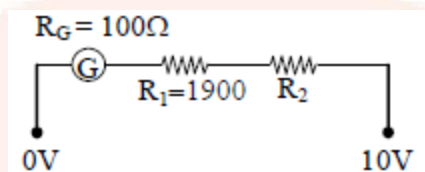
Now, for R_1 , it should be converted in to 2 voltmeter

$$I_{\max} = \frac{2}{100 + R_1} = 10^{-3}$$

$$\Rightarrow \frac{2}{100 + R_1} = 10^{-3}$$

$$\Rightarrow R_1 = 2000 - 100$$

$$\Rightarrow R_1 = 1900 \Omega$$



for R_2 , it should be converted into 10V voltmeter.

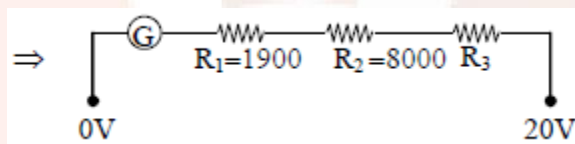
$$I_{\max} = \frac{10}{R_G + R_1 + R_2}$$

$$= 10^{-3} = \frac{10}{100 + 1900 + R_2}$$

$$\Rightarrow R_2 + 2000 = \frac{10}{10^{-3}}$$

$$\Rightarrow R_2 = 10000 - 2000 = 8000 \Omega$$

$$R_G = 100 \Omega$$



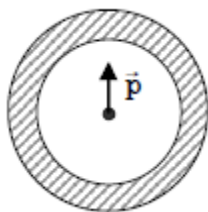
for R_3 , it should be converted into 20V voltmeter.

$$I_{\max} = \frac{20}{100 + 1900 + 8000 + R_3}$$

$$10000 + R_3 = 20000$$

$$R_3 = 10000 \Omega$$

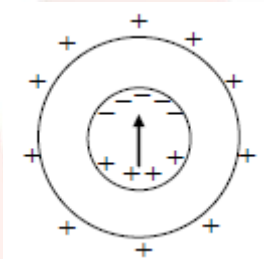
12. Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b , and carries charge Q . At its center $\rightarrow$ is a dipole P as shown. In this case:



- A. electric field outside the shell is the same as that of a point charge at the center of the shell.
- B. Surface charge density on the inner surface is uniform and equal to $\frac{(Q/2)}{4\pi a^2}$
- C. Surface charge density on the inner surface of the shell is zero everywhere.
- D. Surface charge density on the outer surface depends on $|\vec{p}|$.

Ans: A

The charge distribution at equilibrium on the conductor will be like :



Since dipole is having zero net charge. So, Total charge on the inner surface is zero. So, the resultant electric field outside the region would be same as that would have been for point charge at surface.

13. The value of numerical aperture of the objective lens of a microscope is 1.25. If light of wavelength 5000 Å is used, the minimum separation between two points, to be seen as distinct, will be :
- A. 0.49 μm
 - B. 0.12 μm
 - C. 0.24 μm
 - D. 0.38 μm

Ans: C

It is given that Numerical aperture of the microscope is

$$NA = \frac{0.61\lambda}{d}$$

We know that,

d = minimum separation between two points to be seen as distinct

Therefore,

$$\begin{aligned}\Rightarrow d &= \frac{0.61\lambda}{NA} \\ &= \frac{(0.61)(5000 \times 10^{-10})}{1.25} \\ &= 2.4 \times 10^{-7} \text{ m} \\ &= 0.24 \mu\text{m}\end{aligned}$$

Thus, the minimum separation between two points, to be seen as distinct, will be $0.24 \mu\text{m}$.

14. A thin ring of 10 cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of $40 \pi \text{ rad s}^{-1}$ about its axis, perpendicular to its plane. If the magnetic field at its center is $3.8 \times 10^{-9} \text{ T}$, then the charge carried by the ring is close to ($\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$).

- | | |
|---------------------------------|---------------------------------|
| A. $2 \times 10^{-6} \text{ C}$ | B. $4 \times 10^{-5} \text{ C}$ |
| C. $7 \times 10^{-6} \text{ C}$ | D. $3 \times 10^{-5} \text{ C}$ |

Ans: D

Given,

$$\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

$$\omega = 40 \text{ rad/s}$$

$$B_{\text{at centre}} = 3.0 \times 10^{-9} \text{ T}$$

$$R = 10 \text{ cm} = 0.1 \text{ m}$$

Here, I can be determined by flow of charge per rotation

$$I = \frac{\Delta Q}{\Delta t} = \frac{Q}{\left[\frac{2\pi}{\omega} \right]}$$

We know that magnetic field at the centre of a current carrying ring is given by

$$B_0 = \frac{\mu_0 I}{2R} = \frac{\mu_0 \left[\frac{Q}{\left(\frac{2\pi}{\omega} \right)} \right]}{2R}$$

$$\Rightarrow B_0 = \frac{\mu_0 Q \omega}{4\pi R}$$

$$\Rightarrow Q = \frac{B_0 4\pi R}{\mu_0 \omega}$$

$$\begin{aligned} &= \frac{(3.8 \times 10^{-9})(4\pi \times 10^{-1})}{(4\pi \times 10^{-7})(40\pi)} \\ &= \frac{380 \times 10^{-2} \times 10^{-9} \times 10^{-1}}{40 \times 3.14 \times 10^{-7}} \\ &= \frac{380}{125.6} \times 10^{-12+7} = 3 \times 10^{-5} \text{ C} \end{aligned}$$

15. A submarine (A) travelling at 18 km/hr is being chased along the line of its velocity by another submarine (B) travelling at 27 km/hr. B sends a sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν is close to :

(Speed of sound in water = 1500 ms^{-1})

- A. 499 Hz B. 507 Hz
C. 502 Hz D. 504 Hz

Ans: C

Given, velocity of submarine

$$V_A = 18 \text{ km / h} = \frac{1800}{3600} \text{ m / s} = 5 \text{ m / s}$$

and velocity of submarine B

$$V_B = 27 \text{ Km / h} = \frac{27000}{3600} \text{ m / s}$$

$$= 7.5 \text{ m/s}$$

v = speed of sound in water = 1500 m/s

$$\text{frequency received by A} = f' = \left[\frac{v - V_A}{v - V_B} \right] f_0 = \left[\frac{1500 - 5}{1500 - 7.5} \right] f_0$$

$$\text{frequency received by B} = f'' = \left[\frac{v + V_B}{v + V_A} \right] f' = \left[\frac{1500 + 7.5}{1500 + 5} \right] \left[\frac{1500 - 5}{1500 - 7.5} \right] f$$

$$f' = \left(\frac{1500 + 7.5}{1500 - 7.5} \right) \left(\frac{1500 - 5}{1500 + 5} \right) (500)$$

$$= 502 \text{ Hz}$$

16. A point dipole $\vec{P} = -p_0 \hat{x}$ is kept at the origin. The Potential and electric field due to this dipole on the y -axis at a distance d are, respectively: (Take $V = 0$ at infinity)

A. $0, \frac{-\vec{P}}{4\pi \epsilon_0 d^3},$

B. $0, \frac{\vec{P}}{4\pi \epsilon_0 d^3}$

C. $\frac{|\vec{P}|}{4\pi \epsilon_0 d^2}, \frac{-\vec{P}}{4\pi \epsilon_0 d^3}$

D. $\frac{|\vec{P}|}{4\pi \epsilon_0 d^2}, \frac{\vec{P}}{4\pi \epsilon_0 d^3}$

Ans: A

A is an equatorial point w.r.t the dipole:

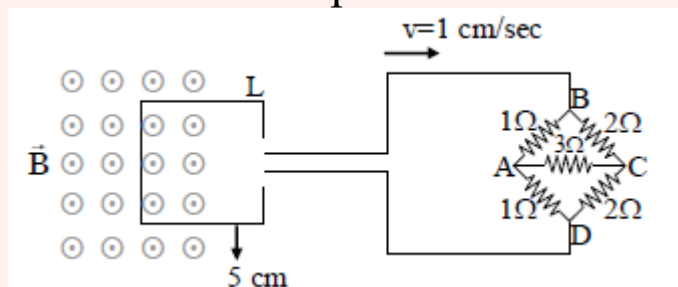
$$\text{Electric field at A} = \frac{-K\vec{p}}{r^3}$$

$$= - \left\{ \frac{1}{4\pi\epsilon_0} \left(\frac{-p_0 \hat{x}}{d^3} \right) \right\}$$

$$= \frac{p_0}{4\pi\epsilon_0 d^3} \hat{x}$$

Electric potential at A = 0

17. The figure shows a square loop L of side 5 cm which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of 1 cm s^{-1} . At some instant, a part of L is in a uniform magnetic field of 1 T, perpendicular to the plane of the loop. If the resistance of L is 1.7Ω , the current in the loop at that instant will be close to:



- A. $115 \mu\text{A}$
C. $60 \mu\text{A}$

- B. $170 \mu\text{A}$
D. $150 \mu\text{A}$

Ans: B

Given that,

$$V = 1 \text{ cm/s} = 10^{-2} \text{ m/s}$$

$$R = 1.7 \Omega$$

$$A = 5 \text{ cm} = 5 \times 10^{-2} \text{ m equivalent circuit : } R_{\text{eq}} = R_l + R_w$$

$$R_w = \frac{(4)(2)}{4+2} = \frac{8}{6} = \frac{4}{3} = 1.3 \Omega$$

Now,

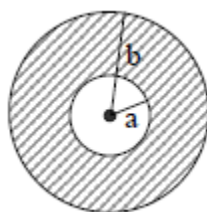
$$R_{\text{eq}} = 1.7 + 1.3$$

$$\Rightarrow R_{\text{eq}} = 3 \Omega$$

Induced emf in the conductor of length l moving with velocity of 1 m/s in the magnetic field 1 T is given by $= VB\ell$

$$\begin{aligned}\Rightarrow \text{current in the loop} = I &= \frac{(VB\ell)}{R_{eq}} \\ &= \frac{(10^{-2})(1)(5 \times 10^{-2})}{3} \\ &= \frac{5}{3} \times 10^{-4} \\ &= 1.67 \times 10^{-4} \\ &= 1.67 \times 10^{-6} \text{A} \\ &\approx 170 \mu\text{A}\end{aligned}$$

18. A circular disc of radius b has hole of radius a at its center (see figure). If the mass per unit area of the disc varies as $\left(\frac{\sigma_0}{r}\right)$, then the radius of gyration of the disc about its axis passing through the centre is :



A. $\frac{a+b}{2}$

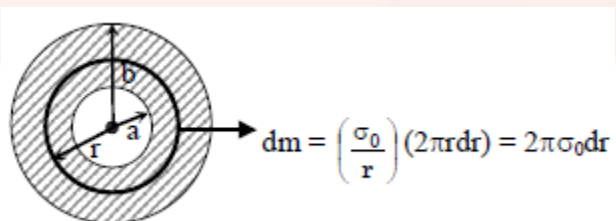
B. $\sqrt{\frac{a^2 + b^2 + ab}{2}}$

C. $\sqrt{\frac{a^2 + b^2 + ab}{3}}$

D. $\frac{a+b}{3}$

Ans: C

Given, mass per unit area $\sigma = \frac{\sigma_0}{r}$



Radius of gyration K of any structure is given by mass of this element is

$$d_m = \sigma \cdot dA = \frac{\sigma_0}{r} \times 2\pi r dr$$

$I = mk^2$: k = radius of gyration

$$\int_a^b (dm)r^2 = k^2 \int_a^b dm$$

$$\Rightarrow \int_a^b (2\pi\sigma_0 dr)r^2 = k^2 \int_a^b 2\pi\sigma_0 dr$$

$$\Rightarrow 2\pi\sigma_0 \left[\frac{b^3 - a^3}{3} \right] = k^2 2\pi\sigma_0 (b - a)$$

$$\Rightarrow \frac{(b - a)(b^2 + a^2 + ab)}{3} = k^2 (b - a)$$

$$k = \sqrt{\frac{a^2 + b^2 + ab}{3}}$$

19. An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n , corresponding to its initial excited state is (for photon of wavelength λ , energy

$$E = \frac{1240\text{eV}}{\lambda(\text{in nm})} :$$

A. $n = 7$

B. $n = 4$

C. $n = 5$

D. $n = 6$

Ans: C

Change in the energy in transition from n to m stage is given by ($n > m$)

$$E_n = +13.6(Z)^2 \left[\frac{1}{m^2} - \frac{1}{n^2} \right] = \frac{hc}{\lambda} \dots (i)$$

Let it start from n to m and from m to ground.

Then, first case, $13.6 \times 4 \left(\frac{1}{m^2} - \frac{1}{n^2} \right) = \frac{hc}{108.5 \text{ nm}} \dots (2)$

Second case, $13.6 \times 4 \left[\frac{1}{1} - \frac{1}{m^2} \right] = \frac{hc}{30.4 \text{ nm}} \dots (3)$

Now,

$$\left(1 - \frac{1}{m^2}\right) = \frac{hc}{30.4 \times 4 \times 13.6} = \frac{1240 \text{ eV}}{30.4 \times 4 \times 13.6}$$

$$1 - \frac{1}{m^2} = 0.75$$

$$\frac{1}{m^2} = 0.25 \Rightarrow m = 4$$

So, by putting the value of m in equation (2), we get $n = 5$

20. When M_1 gram of ice at -10°C (specific heat = $0.5 \text{ cal g}^{-1} ^\circ\text{C}^{-1}$) is added to M_2 gram of water at 50°C , finally no ice is left and the water is at 0°C . The value of latent heat of ice, in cal g^{-1} is :

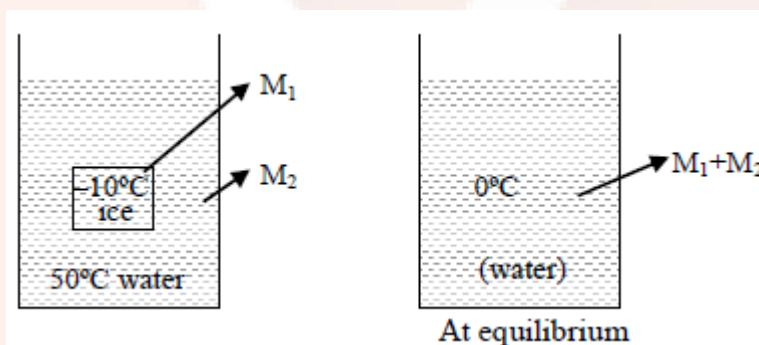
A. $\frac{50M_2}{M_1}$

B. $\frac{5M_1}{M_2} - 50$

C. $\frac{5M_2}{M_1} - 5$

D. $\frac{50M_2}{M_1} - 5$

Ans: D



By using energy conservation law, we get,

$$E_{\text{released by water}} = E_{\text{used by ice}}$$

$$\Rightarrow M_2 S_w (\Delta T)_{\text{water}} = M_1 S_{\text{ice}} (\Delta T)_{\text{ice}} + M_1 L_{\text{fusion}}$$

$$\Rightarrow M_2 (1)(50) = M_1 \left(\frac{1}{2} \right) (10) + M_1 L_{\text{fusion}}$$

$$\Rightarrow 50 M_2 - 5 M_1 = M_1 L_{\text{fusion}}$$

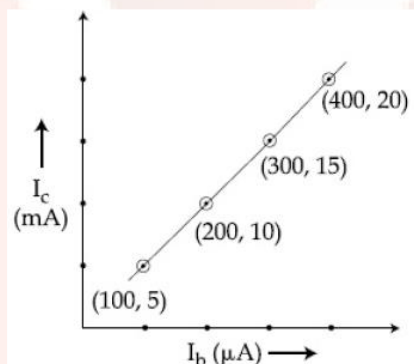
$$\Rightarrow L_f = \frac{50 M_2 + 5 M_1}{M_1}$$

$$\Rightarrow L_f = \frac{50}{M_1} - 5$$

Thus,

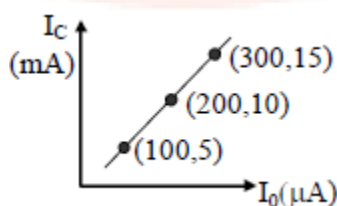
The value of latent heat of ice, in cal g^{-1} is $\frac{50 M_2}{M_1} - 5$.

21. The transfer characteristic curve of a transistor, having input and output resistance 100Ω and $100\text{ K}\Omega$ respectively, is shown in the figure. The Voltage and Power gain, are respectively:



- A. $5 \times 10^4, 5 \times 10^6$
 B. $5 \times 10^4, 2.5 \times 10^6$
 C. $5 \times 10^4, 5 \times 10^5$
 D. $2.5 \times 10^4, 2.5 \times 10^6$

Ans: B



$$B = \frac{I_C}{I_B} = \frac{5 \times 10^{-3}}{100 \times 10^{-9}} = 50$$

Let A_v denote the Voltage gain, then, we get,

$$\text{Voltage gain} = A_v = \frac{V_{\text{output}}}{V_{\text{input}}}$$

$$= \frac{I_{\text{output}} R_0}{I_{\text{input}} R_{\text{in}}}$$

$$= \left(\frac{I_C}{I_b} \right) \left(\frac{R_0}{R_{\text{in}}} \right)$$

$$= (\beta) \left(\frac{R_0}{R_{\text{in}}} \right)$$

$$= 50 \left[\frac{100 \times 10^3}{100} \right]$$

$$= 5 \times 10^4$$

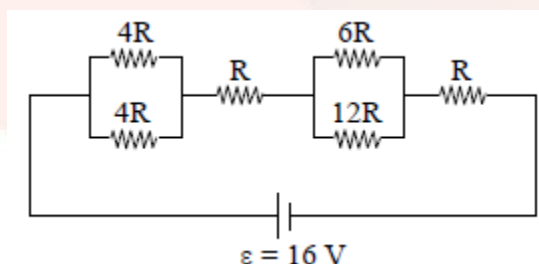
$$\text{Power gain} = \beta^2 = \left(\frac{R_0}{R_i} \right)$$

$$= (50)^2 \times 10^3$$

$$= 25 \times 10^2 \times 10^3$$

$$= 2.5 \times 10^6$$

22. The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is:



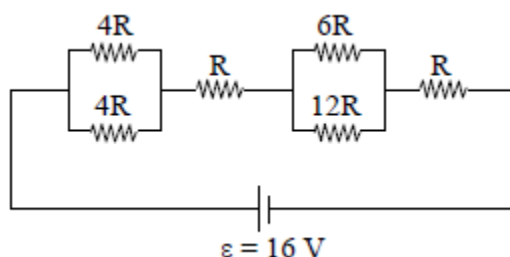
A. 6Ω

B. 1Ω

C. 16Ω

D. 8Ω

Ans: D



$$R_{eq} = 2R + R + 4R = 8R$$

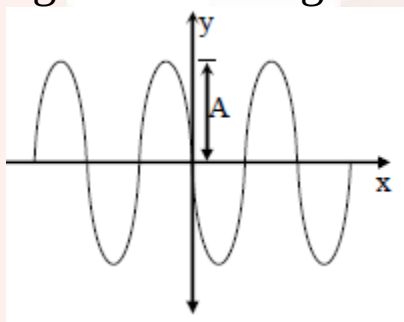
Power consumed by the network

$$P = \frac{V^2}{R}$$

$$\Rightarrow 4 = \frac{16 \times 16}{8R}$$

$$\Rightarrow \boxed{R = 8\Omega}$$

23. A progressive wave travelling along the positive x – direction is represented by $(x, t) = A \sin (kx - \omega t + \phi)$. Its snapshot at $t = 0$ is given in the figure.



For this wave, the phase ϕ is :

- A. $-\frac{\pi}{2}$ B. 0
C. $\frac{\pi}{2}$ D. π

Ans: D

$$Y = A \sin (kx - \omega t + \phi)$$

At $t = 0$ and $x_0 = 0$ particle is at mean position and will proceed in positive y direction.

$$\begin{aligned}
 -A \sin(kx) &= A \sin(kx + \phi) \\
 \Rightarrow A \sin(kx + \pi) &= A \sin(kx + \phi) \\
 \Rightarrow \phi &= \pi
 \end{aligned}$$

24. A magnetic compass needle oscillates 30 times per minute at a place where the dip is 45° , and 40 times per minute where the dip is 30° . If B_1 and B_2 are respectively the total magnetic field due to the earth at the two places, then the ratio B_1/B_2 is best given by:

- A. 2.2 B. 1.8
C. 3.6 D. 0.7

Ans: D

Given, at first place, angle of dip $\theta_1 = 45^\circ$

$$\text{So, } T_1 = \frac{60}{30} = 2\text{S}$$

$$B_{H_1} = B_1 \cos 45^\circ = \frac{B_1}{\tau_2}$$

At second place, angle of dip $\theta_2 = 30^\circ$

$$T_2 = \frac{60}{40} = \frac{3}{2}\text{S}$$

$$B_{H_2} = B_2 \cos 30^\circ = \frac{B_2 \sqrt{3}}{2}$$

Time period of a magnetic needle is given by

$$T = 2\pi \sqrt{\frac{I}{MB_H}}$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{B_{H_2}}{B_H}}$$

After putting value we get

$$\frac{B_1}{B_2} = 0.7$$

25. In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the center of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used):

- A. $\frac{2\lambda}{(\mu - 1)}$ B. $\frac{\lambda}{(2\mu - 1)}$
C. $\frac{\lambda}{2(\mu - 1)}$ D. $\frac{\lambda}{(\mu - 1)}$

Ans: D

For central bright at the position of first bright

$$\Rightarrow S_2P - S_1P = 0$$

$$\Rightarrow (\lambda + \ell) - (\ell - t + \mu t) = 0$$

optical path length

$$\Rightarrow \lambda + \ell - \ell + t - \mu t = 0$$

$$\Rightarrow \lambda = t(\mu - 1)$$

$$\Rightarrow \boxed{t = \frac{\lambda}{\mu - 1}}$$

26. A person of mass M is, sitting on a swing of length L and swinging with an angular amplitude θ_0 . If the person through its lowest point, the work done by him, assuming that his centre of mass moves by a distance l ($l \ll L$), is close to:

- A. $Mgl(1 + \theta_0^2)$ B. Mgl
C. $Mgl(1 + \frac{\theta_0^2}{2})$ D. $Mgl(1 - \theta_0^2)$

Ans: A

$$W_{\text{man}} = M g_{\text{eff}} l$$

$$g_{\text{eff}} = g(1 + \theta_0^2)$$

$$W_{\text{man}} = Mgl(1 + \theta_0^2)$$

27. An electromagnetic wave is represented by the electric field $\vec{E} = E_0 \sin[\omega t + (6y - 8z)]$. Taking unit vectors in $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Z}$ directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propagation, $\hat{s}$, is :

A. $\hat{s} = \frac{4\hat{j} - 3\hat{k}}{5}$

B. $\hat{s} = \frac{-4\hat{k} + 3\hat{j}}{5}$

C. $\hat{s} = \frac{-3\hat{j} + 4\hat{k}}{5}$

D. $\hat{s} = \frac{3\hat{i} - 4\hat{j}}{5}$

Ans: Given that

An electromagnetic wave

$$\vec{E} = E_0 \hat{n} [\omega t + (6y - 8z)]$$

$$\Rightarrow \vec{E} = E_0 \hat{n} [\omega t - (8z - 6y)] \dots (1)$$

Standard expression of E_m waves is given by

$$E = E_0 \hat{n} [\sin(\omega t - \mathbf{K} \cdot \hat{r})]$$

Here, $\mathbf{K}$ is the propagation vector

$$\Rightarrow \vec{E} = E_0 \hat{n} [\omega t - \left[\frac{8}{10} \hat{k} - \frac{6}{10} \hat{j} \right] \cdot 10]$$

$$\Rightarrow \vec{E} = E_0 \hat{n} [\omega t - \hat{s} k]$$

$\hat{s}$ = direction of propagation

$$\hat{s} = \hat{K} = \frac{\mathbf{K}}{|\mathbf{K}|} = \frac{-6\hat{j} + 8\hat{k}}{\sqrt{6^2 + 8^2}} = \frac{-6\hat{j} + 8\hat{k}}{10}$$

$$\hat{s} = \left(\frac{8\hat{k} - 6\hat{j}}{10} \right)$$

$$= \left(\frac{4\hat{k} - 3\hat{j}}{5} \right)$$

$$= \frac{-3\hat{j} + 4\hat{k}}{5}$$

28. At 40°C , a brass wire of 1 mm radius is hung from the free end of the wire. When the wire is cooled down from 40°C to 20°C it regains its original length of 0.2 m. The value of M is close to:

(Coefficient of linear expansion and Young's modulus of brass are $10^{-5}/^{\circ}\text{C}$ and 10^{11} N/m^2 , respectively; $g = 10 \text{ ms}^{-2}$)

- A. 9 kg B. 1.5 kg
C. 0.9 kg D. 0.5 kg

Ans: A

Given that

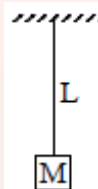
$$r = 1 \text{ mm} = 10^{-3} \text{ m}$$

$$L_0 = 0.2 \text{ m}$$

$$T_1 = 40^{\circ}\text{C} \text{ and } T_2 = 20^{\circ}\text{C} \Rightarrow \Delta T = T_1 - T_2 = 20^{\circ}\text{C}$$

$$\text{Young's modulus, } Y = 10^{11} \text{ N/m}^2$$

$$\text{linear expansion } \alpha = 10^{-5}/^{\circ}\text{C}$$



$$\Delta L = L_0 \times \Delta T$$

$$Y = \frac{(F / A)}{(\Delta L / L_0)}$$

$$\Rightarrow \frac{\Delta L}{L_0} = \frac{F}{AY}$$

$$\Rightarrow \Delta L = \frac{FL_0}{AY}$$

$$\Rightarrow L_0 \alpha \Delta T = \frac{FL_0}{AY}$$

$$\Rightarrow \frac{MgL_0}{AY} = L_0 \alpha \Delta T$$

$$\Rightarrow M = \frac{YA}{g} \times \alpha \Delta T = \frac{10^{11} \times 22 \times 10^{-6} \times 10^{-5} \times 20}{7 \times 10}$$

$$= 6.28 \text{ kg}$$

29. Two moles of helium gas is mixed with three moles of hydrogen molecules (taken to be rigid). What is the molar specific heat of mixture at constant volume?

($R = 8.3 \text{ J/mol K}$)

- A. 17.4 J/mol K B. 21.6 J/mol K
C. 15.7 J/mol K D. 19.7 J/mol K

Ans: A

It is given that two moles of helium gas is mixed with three moles of hydrogen molecules,

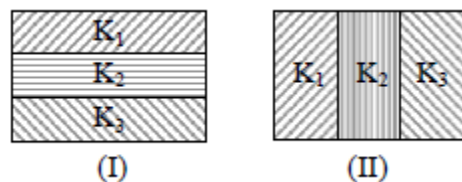
So, according to the given conditions, we have,

$$\begin{aligned}(C_V)_{\max} &= \frac{n_1 C_{V_1} + n_2 C_{V_2}}{n_1 + n_2} \\&= \frac{(2) \left[\frac{fR}{2} \right]_{\text{mono}} + 3 \left[\frac{fR}{2} \right]_{\text{diatomic}}}{2 + 3} \\&= \frac{2 \left(\frac{3R}{2} \right) + 3 \left(\frac{5R}{2} \right)}{5} \\&= \frac{21R}{10} \\&= \frac{21 \times 8.3}{10} = 17.4 \text{ J/mol-K}\end{aligned}$$

Hence, 17.4 J/mol K is the molar specific heat of mixture at constant volume.

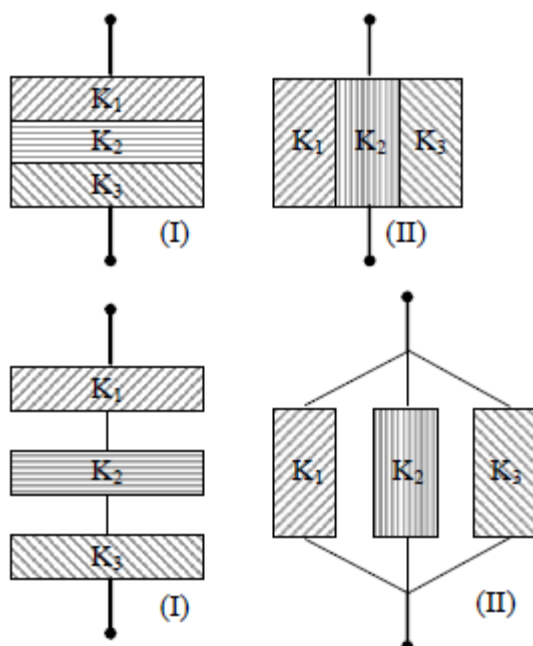
30. Two identical parallel plate capacitors, of capacitance C each, have plates of area A , separated by a distance d . The space between the plates of the two capacitors, is filled with three dielectric constants K_1 , K_2 and K_3 . The first capacitor is filled as shown in fig. I, and the second one is filled as shown in fig II.

If these two modified capacitors are charged by the same potential V , the ratio of the energy stored in the two, would be (E_1 refers to capacitor (I) and E_2 to capacitor (II)):



- A. $\frac{E_1}{E_2} = \frac{9K_1K_2K_3}{(K_1 + K_2 + K_3)(K_2K_3 + K_3K_1 + K_1K_2)}$
- B. $\frac{E_1}{E_2} = \frac{K_1K_2K_3}{(K_1 + K_2 + K_3)(K_2K_3 + K_3K_1 + K_1K_2)}$
- C. $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2K_3 + K_3K_1 + K_1K_2)}{9K_1K_2K_3}$
- D. $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2K_3 + K_3K_1 + K_1K_2)}{K_1K_2K_3}$

Ans: A



$$\frac{1}{C_1} = \frac{1}{k_1 \left(\frac{\epsilon_0 A}{d/3} \right)} + \frac{1}{k_2 \left(\frac{\epsilon_0 A}{d/3} \right)} + \frac{1}{k_3 \left(\frac{\epsilon_0 A}{d/3} \right)}$$

$$\frac{1}{C_1} = \frac{d}{3\epsilon_0 A} \left[\frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} \right]$$

$$C_1 = \frac{3\epsilon_0 A}{d} \frac{k_1 k_2 k_3}{(k_1 k_2 + k_2 k_3 + k_3 k_1)}$$

$$C_2 = k_1 \frac{\epsilon_0 A / 3}{d} + k_2 \frac{\epsilon_0 A / 3}{d} + k_3 \frac{\epsilon_0 A / 3}{3d} = \frac{\epsilon_0 A}{3d} [k_1 + k_2 + k_3]$$

So, the ratio of the energy stored in the two, would be (E_1 refers to capacitor (I) and E_2 to capacitor (II)) is

$$\frac{E_1}{E_2} = \frac{\frac{1}{2} C_1 V^2}{\frac{1}{2} C_2 V^2} = \frac{9 k_1 k_2 k_3}{(k_1 + k_2 + k_3)(k_1 k_2 + k_2 k_3 + k_3 k_1)}$$

Chemistry

31. 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of A (M_A) and molar mass of B (M_B) in kg mol^{-1} are:

- A. $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$
- B. $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$
- C. $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$
- D. $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$

Ans: C

Given that, 5 mol AB_2 weigh 125 g

$$\text{Number of moles} = \frac{\text{given mass}}{\text{molar mass}}$$

$$5 = \frac{125}{M_A + 2M_B}$$

$$M_A + 2M_B = 25 \quad \dots(i)$$

10 mol A_2B_2 weighs 300 g

$$10 = \frac{300}{2M_A + 2M_B}$$

$$2M_A + 2M_B = 30 \dots(2)$$

Solving (1) & (2) ;

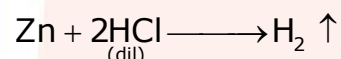
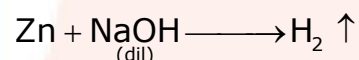
$$M_A = 5 \text{ g/mol} = 5 \times 10^{-3} \text{ kg/mol}$$

$$M_B = 10 \text{ g/mol} = 10 \times 10^{-3} \text{ kg/mol}$$

32. The metal that gives hydrogen gas upon treatment with both acid as well as base is:

- | | |
|--------------|---------|
| A. Magnesium | B. Zinc |
| C. Mercury | D. Iron |

Ans: B



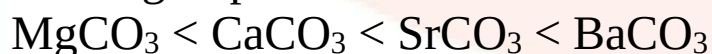
Zn is amphoteric.

33. The correct sequence of thermal stability of the following carbonates is:

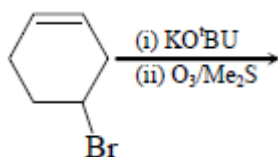
- $\text{MgCO}_3 < \text{CaCO}_3 < \text{SrCO}_3 < \text{BaCO}_3$
- $\text{BaCO}_3 < \text{SrCO}_3 < \text{CaCO}_3 < \text{MgCO}_3$
- $\text{MgCO}_3 < \text{SrCO}_3 < \text{CaCO}_3 < \text{BaCO}_3$
- $\text{BaCO}_3 < \text{CaCO}_3 < \text{SrCO}_3 < \text{MgCO}_3$

Ans: A

In 2nd group carbonates thermal stability increases down the group.



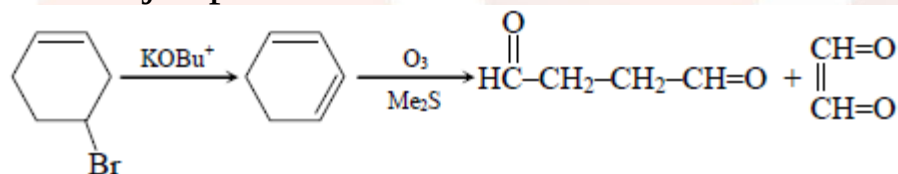
34. The major product (s) in the following reaction is/are :



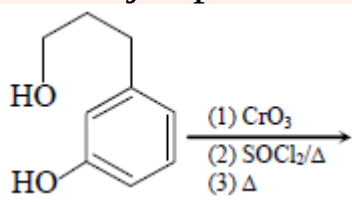
- A. O=CCCC=CC=O
- B. O=CC(OtBu)CC=O
- C. O=CC=O
- D. O=CC=O and O=CC=O

Ans: D

The major product formed in the reaction is as follows:



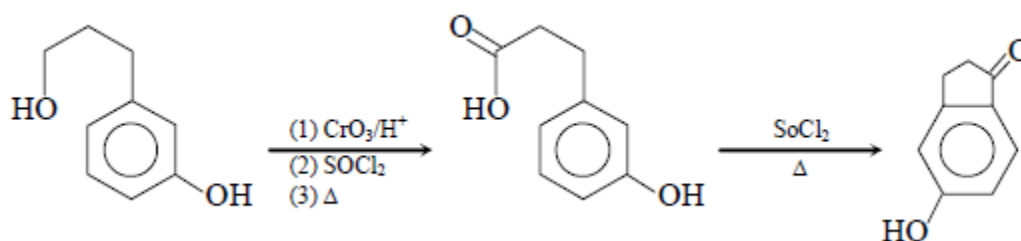
35. The major product of the following reaction is:



- A.
- B.
- C.
- D.

Ans: A

The major product formed in the reaction is as follows:



36. An element has a face-centred cubic (fcc) structure with a cell edge of a . The distance between the centres of two nearest tetrahedral voids in the lattice is:

- A. $\frac{3}{2}a$ B. a
C. $\frac{a}{2}$ D. $\sqrt{2}a$

Ans: C

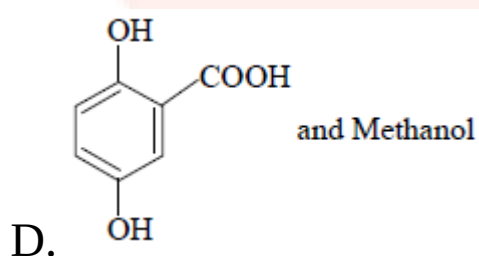
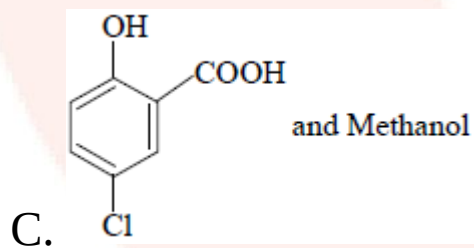
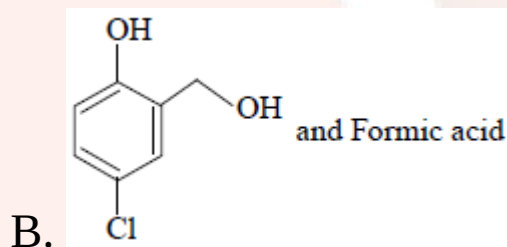
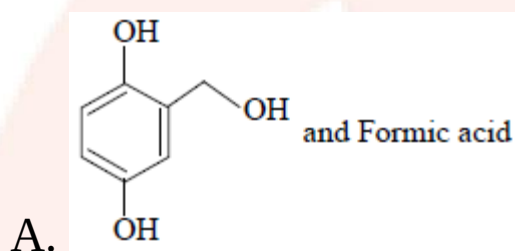
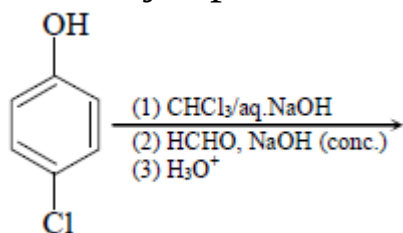
In fcc, tetrahedral voids are located on the body diagonal at a distance of $\frac{\sqrt{3}a}{4}$ from the corner. Minimum distance between two tetrahedral voids is $\frac{a}{2}$.

37. Complete removal of both the axial ligands (Along the z -axis) from an octahedral complex leads to which of the following splitting patterns? (relative orbital energies not on scale).

- A.
- B.
- C.
- D.

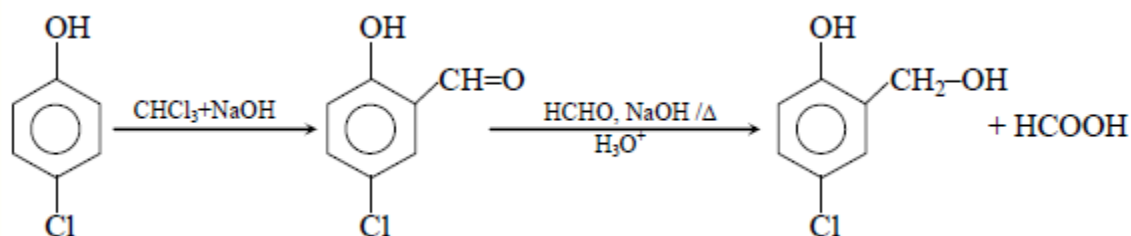
Ans: A

38. The major products of the following reaction are :

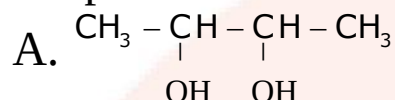


Ans: B

The major product formed in the reaction is as follows:



39. But – 2- ene on reaction with alkaline KMnO_4 at elevated temperature followed by acidification will give :



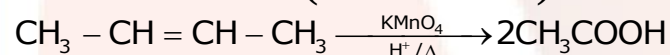
B. 2 molecules of CH_3COOH

C. One molecule of CH_3CHO and one molecule of CH_3COOH

D. 2 molecules of CH_3CHO

Ans: B

But-2-ene on reaction with alkaline KMnO_4 acidification will give acetic acid (CH_3COOH).



40. What is the molar solubility $\text{Al}(\text{OH})_3$ in 0.2 M NaOH solution ? Given that, solubility product of $\text{Al}(\text{OH})_3 = 2.4 \times 10^{-24}$:

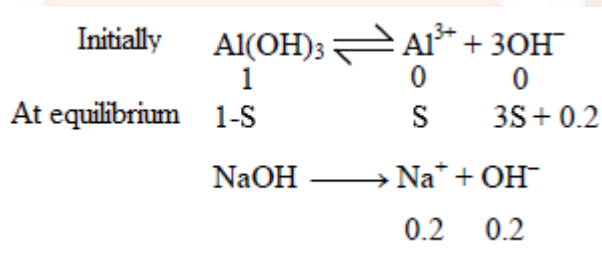
A. 3×10^{-22}

B. 12×10^{-23}

C. 12×10^{-21}

D. 3×10^{-19}

Ans: A



$$K_{\text{sp}} \text{ of } \text{Al}(\text{OH})_3 = 2.4 \times 10^{-24} \text{ (Given)}$$

$$[\text{Al}^{3+}][\text{OH}^-]^3 = K_{\text{sp}}$$

$$S(3S + 0.2)^3 = K_{\text{sp}} \quad \because 0.2 \gg 5$$

$$S \times 0.008 = 2.4 \times 10^{-24}$$

$$S = 3 \times 10^{-22}$$

41. The group number, number of valence electrons, and valency of an element with atomic number 15, respectively, are :

- A. 16, 6 and 3 B. 15, 5 and 3
C. 15, 6 and 2 D. 16, 5 and 2

Ans: B

Given that

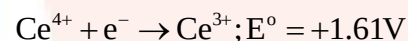
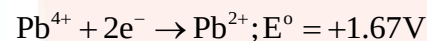
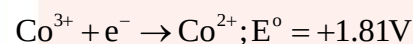
Atomic number $z = 15$ ($1s^2 2s^2 2p^6 3s^2 3p^3$)

Electronic configuration of element

Valence e^- s = 5

Valency = 3

42. Given :

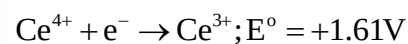
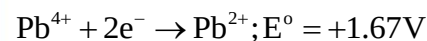
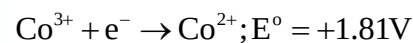


Oxidizing power of the species will increase in the order :

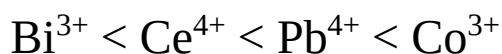
- A. $\text{Ce}^{4+} < \text{Pb}^{4+} < \text{Bi}^{3+} < \text{Co}^{3+}$
B. $\text{Co}^{3+} < \text{Ce}^{4+} < \text{Bi}^{3+} < \text{Pb}^{4+}$
C. $\text{Bi}^{3+} < \text{Ce}^{4+} < \text{Pb}^{4+} < \text{Co}^{3+}$
D. $\text{Co}^{3+} < \text{Pb}^{4+} < \text{Ce}^{4+} < \text{Bi}^{3+}$

Ans: C

Given,



Oxidising power of the species increases in the order.



43. Glucose and Galactose are having identical configuration in all the positions except position.

- A. C – 4. B. C – 2.
C. C – 5. D. C – 3.

Ans: A

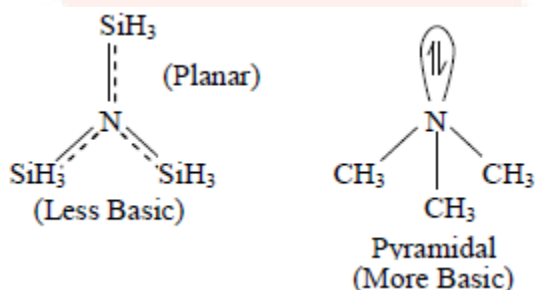
Glucose and galactose are having identical configuration at all the positions except C_4 .

44. The correct statement among the following is :

- A. $(\text{SiH}_3)_3\text{N}$ is planar and less basic than $(\text{CH}_3)_3\text{N}$.
B. $(\text{SiH}_3)_3\text{N}$ is planar and more basic than $(\text{CH}_3)_3\text{N}$.
C. $(\text{SiH}_3)_3\text{N}$ is pyramidal and more basic than $(\text{CH}_3)_3\text{N}$.
D. $(\text{SiH}_3)_3\text{N}$ is pyramidal and less basic than $(\text{CH}_3)_3\text{N}$.

Ans: B

The correct statement is that $(\text{SiH}_3)_3\text{N}$ is planar and less basic than $(\text{CH}_3)_3\text{N}$.



45. Peptization is a :

- A. Process of converting precipitate into colloidal solution
B. Process of converting soluble particles to form colloidal solution
C. Process of converting a colloidal molecule into solution
D. process of converting a colloidal solution into precipitate

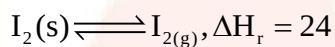
Ans: A

Peptization is a process of converting precipitate into colloidal solution.

46. Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200°C . If specific heat of $\text{I}_2(\text{s})$ and $\text{I}_2(\text{vap})$ are 0.055 and $0.031 \text{ cal g}^{-1} \text{ K}^{-1}$ respectively, then enthalpy of sublimation of iodine at 250°C in cal g^{-1} is :

- A. 22.8 B. 11.4
C. 2.85 D. 5.7

Ans: A



$\Rightarrow$ Specific heat of $\text{I}_2(\text{s}) = 0.055 \text{ cal g}^{-1} \text{ K}^{-1}$

Specific heat of $\text{I}_2(\text{V}_{\text{ap}}) = 0.031 \text{ cal g}^{-1} \text{ K}^{-1}$

As we know that $\Delta H = C_p \Delta T$

$$\Delta H_{rT_2} - \Delta H_{rT_1} = \Delta C_p (T_2 - T_1)$$

$$\Delta H_{r250} = 24 = (0.031 - 0.055) \times 50$$

$$\Delta H_{r250} = 24 - 1.2 = 22.8$$

47. The mole fraction of a solvent in a aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is :

- A. 13.88 B. 13.88×10^{-1}
C. 13.88×10^{-2} D. 13.88×10^{-3}

Ans: A

Given that

The mole fraction of a solvent $n_1 = 0.8$, $n_2 = 0.2$ (Let total 1 moles be present)

$$n_2 = \frac{mM_A}{1000 + mM_A} \text{ or } m = \frac{1000}{M_A} \times \frac{n_2}{1 - n_2}$$

$$\text{Using formula } m = n_2 \times \frac{1000}{n_1 \times M_1}$$

$$m = \frac{1000}{16} \times \frac{0.2}{0.8}$$

$$m = 13.88$$

48. The idea of froth floatation method came from a person X and this method is related to the process Y of ores. X and Y, respectively, are :

- A. Fisher man and reduction
- B. Washer man and reduction
- C. Fisher woman and concentration
- D. Washer woman and concentration

Ans: D

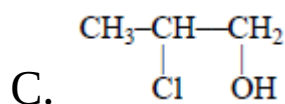
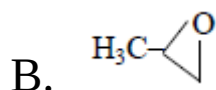
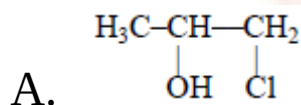
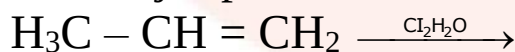
Froth floatation is used for concentration for sulphide ore. Froth floatation is a method of concentration. The idea of froth floatation method came from a person Washer woman.

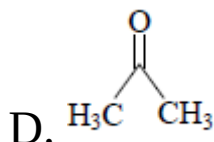
49. The correct set of species responsible for the photochemical smog is :

- A. N_2 , O_2 , O_3 and hydrocarbons
- B. N_2 , NO_2 , and hydrocarbons
- C. NO , NO_2 , O_3 and hydrocarbons
- D. CO_2 , NO_2 , SO_3 and hydrocarbons

Ans: C

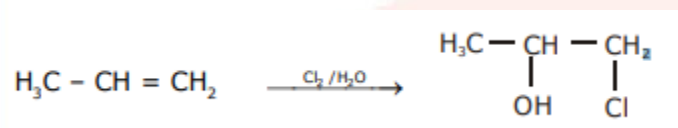
50. The major product of the following addition reaction is





Ans A)

The major product formed in the reaction is as follows



H₂O is used as solvent.

51. In the following reaction; $x\text{A} \rightarrow y\text{B}$ $\log_{10} \left[-\frac{d[\text{A}]}{dt} \right] \log_{10} \left[\frac{d[\text{B}]}{dt} \right] +$

0.3010 'A' and 'B' respectively can be :

A. N₂O₄ and NO₂

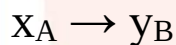
B. C₂H₂ and C₆H₆

C. C₂H₄ and C₄H₈

D. N – Butane and Iso – butane

Ans: C

In the given reaction



$$-\frac{1}{x} \frac{\Delta \text{A}}{dt} = +\frac{1}{y} \frac{\Delta \text{B}}{dt}$$

$$-\frac{\Delta \text{A}}{dt} = +\frac{x}{y} \frac{\Delta \text{B}}{dt} \quad \dots(\text{i})$$

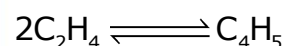
$$\therefore \log \left[-\frac{d\text{A}}{dt} \right] = \log \left[+\frac{d\text{B}}{dt} \right] + \log 2$$

$$-\frac{d\text{A}}{dt} = +\frac{2}{1} \frac{d\text{B}}{dt} \quad \dots(\text{ii})$$

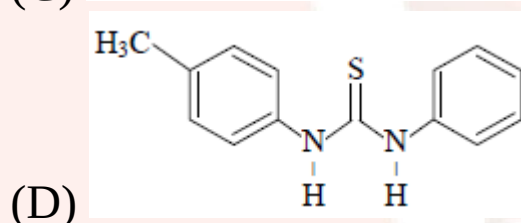
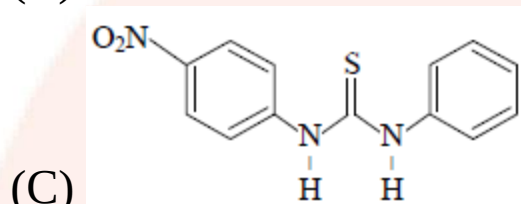
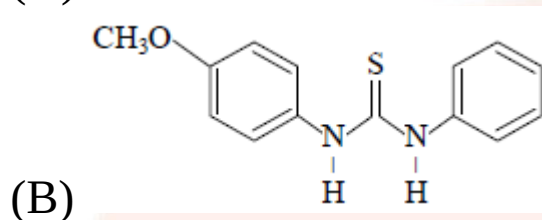
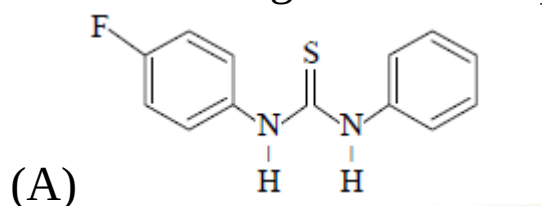
Compare (1) & (2)

$$x = 2, y = 1$$

Using the rate equation to determine the reaction involved is



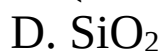
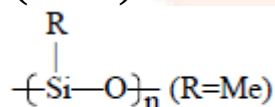
52. The increasing order of the pK_b of the following compound is :



- A. $A < C < D < B$ B. $B < D < C < A$
C. $C < A < D < B$ D. $B < D < A < C$

Ans: D

53. The basic structural unit of feldspar, zeolites, mica, and asbestos is :



Ans: A

The basic structural unit of Feldspar, zeolites, mica and asbestos contains $(\text{SiO}_4)^{4-}$ basic unit.

54. An ideal gas is allowed to expand from 1 L to 10 L against a constant external pressure of 1 bar. The work done in kJ is :

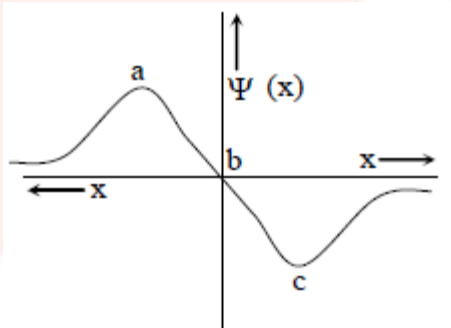
- A. -0.9 B. -9.0
C. $+10.0$ D. -2.0

Ans: A

Work done during isothermal expansion of an ideal is given by

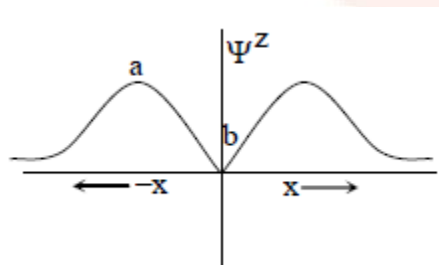
$$\begin{aligned} w &= -P_{\text{ext}} (V_2 - V_1) \\ &= -1 (10 - 1) \\ &= -9 \text{ L-bar} \\ &= -9 \times 100 \text{ J} = -0.9 \text{ kJ} \quad (\because 1 \text{ L bar} = 100 \text{ J}) \end{aligned}$$

55. The electrons are more likely to be found:



- A. Only in the region c B. In the region a and c
C. In the region a and b D. Only in the region a

Ans: B



The electrons are more likely to be found in the region at a and c.

56. Which of the following is a thermosetting polymer?

- | | |
|-------------|------------|
| A. Bakelite | B. Nylon 6 |
| C. Buna – N | D. PVC |

Ans: A

Bakelite is a thermosetting polymer, Bakelite is a cross linked and branched molecule, which upon heating undergoes extensive cross linking in the molds and becomes unstable again. And it cannot be used again that's why it is thermosetting polymer.

57. Which of the following statements is not true about RNA?

- A. It has always double stranded α - helix structure
- B. It usually does not replicate
- C. It is present in the nucleus of the cell
- D. It controls the synthesis of protein

Ans: A

RNA is single stranded helix structure. It does not have double stranded α -helix structure.

58. An organic compound 'A' is oxidized with N_2O_2 followed by boiling with HNO_3 . The resultant solution is then treated with ammonium molybdate to yield a yellow precipitate.

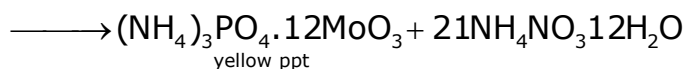
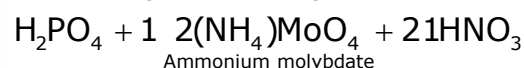
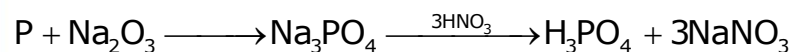
Based on above observation, the element present in the given compound is :

- | | |
|---------------|-------------|
| A. Phosphorus | B. Nitrogen |
| C. Fluorine | D. Sulphur |

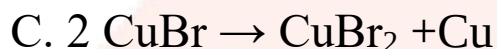
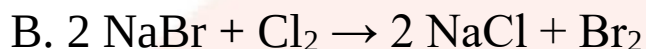
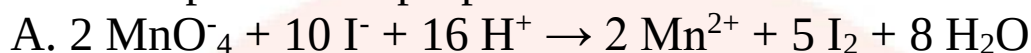
Ans: A

Organic compound A contain phosphorus as it gives positive test with ammonium molybdate.

This is qualitative test for phosphorous

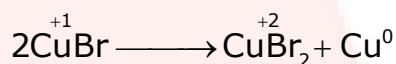


59. An example of a disproportionation reaction is:

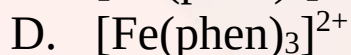
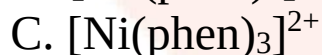
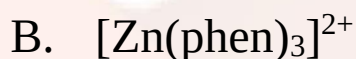
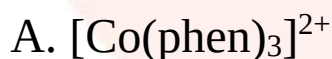
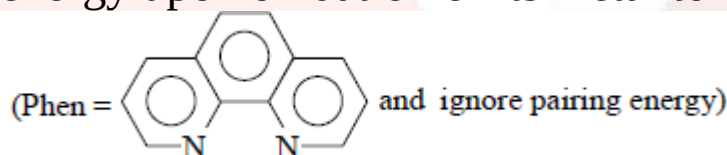


Ans: C

In disproportionation reactions same element undergoes oxidation as well as reduction.



60. The complex ion that will lose its crystal field stabilization energy upon oxidation of its metal to + 3 state is :



Ans: D

The complex ion that will lose its crystal field stabilization energy upon oxidation of its metal to + 3 state is $[Fe(phen)_3]^{2+}$

Mathematics

61. If the area (in sq. units) of the region

$$\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$$

is $a\sqrt{2} + b$, then $a - b$ is equal to :

A. $\frac{10}{3}$

B. $\frac{8}{3}$

C. $-\frac{2}{3}$

D. 6

Ans: D

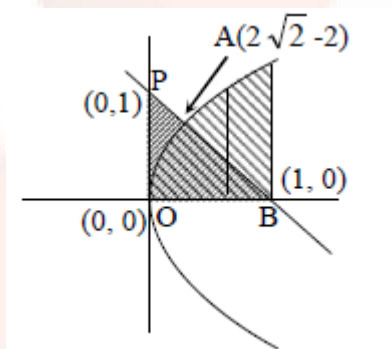
It is given that,

$$C_1 : y^2 \leq 4x$$

$$C_2 : x + y \leq 1$$

$$x \geq 0$$

$$y \geq 0$$



So, now,

$$y^2 = 4x;$$

$$y^2 = 4(1-y) \quad (\text{since, } C_2 : x + y \leq 1)$$

$$y^2 + 4y + 4 = 0$$

So, solving the above equation, we get,

$$y = 2\sqrt{2} - 2, -2\sqrt{2} - 2$$

Now, plotting the points on graph, we get,

The required area is shaded region of curve OAB,

Thus,

$A = \text{Area of } \Delta_{OBP} - \text{Area of region OAP}$

$$\Delta_{OBP} = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$\text{Area of OAP} = \int_0^{2\sqrt{2}-2} \frac{y^2}{4} dy + \int_{2\sqrt{2}-2}^1 (1-y)$$

$$= \frac{1}{12} [y^3]_0^{2\sqrt{2}-2} + \left[y - \frac{y^2}{2} \right]_{2\sqrt{2}-2}^1$$

$$= \frac{1}{12} \left[(2\sqrt{2}-2)^3 \right] + \left[\left(1 - \frac{1}{2} \right) - \left\{ (2\sqrt{2}-2) - \frac{(2\sqrt{2}-2)^2}{2} \right\} \right]$$

$$= \frac{23}{6} - \frac{8}{3}\sqrt{2}$$

$$A = \frac{1}{2} - \frac{23}{6} + \frac{8\sqrt{2}}{3}$$

$$a = \frac{8}{3}, b = -\frac{20}{6}$$

$$\therefore a - b = 6$$

62. If the normal to the ellipse $3x^2 + 4y^2 = 12$ at a point P on it is parallel to the line, $2x + y = 4$ and the tangent to the ellipse at P passes through Q(4,4) then PQ is equal to :

A. $\frac{\sqrt{221}}{2}$

B. $\frac{5\sqrt{5}}{2}$

C. $\frac{\sqrt{157}}{2}$

D. $\frac{\sqrt{61}}{2}$

Ans: B

Equation of given ellipse is $3x^2 + 4y^2 = 12$

$$\frac{x^2}{4} + \frac{y^2}{3} = 1 \quad \dots(1)$$

Now, let point P($2 \cos \theta$, $\sqrt{3} \sin \theta$), so equation of tangent to ellipse 1 at point P is

$$\frac{x \cos \theta}{2} + \frac{y \sin \theta}{\sqrt{3}} = 1 \quad \dots(ii)$$

$$2x \sec \theta - \sqrt{3} \operatorname{cosec} \theta = 1$$

$$\text{Slope of normal} = \frac{-2 \sec \theta}{-\sqrt{3} \operatorname{cosec} \theta} = \frac{2 \sin \theta}{\sqrt{3} \cos \theta}$$

Since, normal parallel to $2x + y = 4$

$\therefore$ slope of normal = slope of line

$$\text{then } \frac{2}{\sqrt{3}} \tan \theta = -2; \tan \theta = -\sqrt{3}$$

$$\theta = \frac{2\pi}{3}$$

$$(\sin \theta, \cos \theta) = \left(\frac{\sqrt{3}}{2}, \frac{-1}{2} \right)$$

Now, point $P\left(-1, \frac{3}{2}\right), Q(4, 4)$

$$PQ = \sqrt{(4+1)^2 + \left(4 - \frac{3}{2}\right)^2} = \sqrt{\frac{125}{4}} = \frac{5\sqrt{5}}{2}$$

63. If three of the six vertices of a regular hexagon are chosen at random, then the probability that the triangle formed with these chosen vertices is equilateral is :

A. $\frac{3}{20}$

B. $\frac{1}{5}$

C. $\frac{1}{10}$

D. $\frac{3}{10}$

Ans: C

Since, there is a regular hexagon, then the number of ways choosing three vertices is 6C_3

Number of total triangle = 6C_3

Equilateral triangle = 2

Required probability = $\frac{2}{{}^6C_3} = \frac{2}{\frac{6!}{3!3!}} = \frac{1}{10}$

64. If the volume of parallelepiped formed by the vectors $\hat{i} + \lambda\hat{j} + \hat{k}$, $\hat{j} + \lambda\hat{k}$ and $\lambda\hat{i} + \hat{k}$ is minimum, then λ is equal to:

- A. $-\sqrt{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. $-\frac{1}{\sqrt{3}}$

Ans: B

Given, vectors are $\hat{i} + \lambda\hat{j} + \hat{k}$, $\hat{j} + \lambda\hat{k}$ and $\lambda\hat{i} + \hat{k}$ which forms a parallelepiped.

$\therefore$ Volume of parallelepiped

$$\begin{vmatrix} 1 & \lambda & 1 \\ 0 & 1 & \lambda \\ \lambda & 0 & 1 \end{vmatrix} = 1[1 - 0] - \lambda[-\lambda^2] + 1[-\lambda]$$

$$V = |1 + \lambda^3 - \lambda|$$

on differentiating w.r.t., λ , we get

$$\frac{dV}{d\lambda} = 3\lambda^2 - 1 \Rightarrow \text{for maxima or minima } \frac{dV}{d\lambda} = 0 \Rightarrow \lambda = \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$$

$$\frac{d^2V}{d\lambda^2} = 6\lambda$$

$$\frac{d^2V}{d\lambda^2} > 0 \text{ at } \lambda = \frac{1}{\sqrt{3}}$$

So, volume minimum for $\lambda = \frac{1}{\sqrt{3}}$

65. Let $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ be two vectors. If a vector perpendicular to both the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ has the magnitude 12 then one such vector is:

A. $4(2\hat{i} + 2\hat{j} - \hat{k})$

B. $4(-2\hat{i} - 2\hat{j} + \hat{k})$

C. $4(2\hat{i} - 2\hat{j} - \hat{k})$

D. $4(2\hat{i} + 2\hat{j} + \hat{k})$

Ans: C

Given vectors are

$$\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\text{and } \vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\text{Now, vectors } \vec{a} + \vec{b} = 4\hat{i} + 4\hat{j}$$

$$\text{and } \vec{a} - \vec{b} = 2\hat{i} + 4\hat{k}$$

Let a vector $\vec{r}$ perpendicular to $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ and magnitude is 12, then

$$\vec{r} = 12.\hat{n}$$

$$\therefore \hat{n} = \frac{(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})}{|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|} \quad \dots(1)$$

$$(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 4 & 0 \\ 2 & 0 & 4 \end{vmatrix}$$

$$= \hat{i}(16) - \hat{j}(16) + \hat{k}(-8) = 8[2\hat{i} - 2\hat{j} - \hat{k}]$$

$$|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})| = 8 \times \sqrt{4 + 4 + 1} = 8 \times 3$$

$$\hat{n} = \frac{(2\hat{i} - 2\hat{j} - \hat{k})}{3}$$

Putting values in eq (1) we get

$$\vec{r} = 4.(2\hat{i} - 2\hat{j} - \hat{k})$$

66. If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$,

then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$ is equal to:

A. $\frac{29}{358}$

B. $\frac{7}{116}$

C. $\frac{1}{12}$

D. $\frac{21}{346}$

Ans: C

Given, α and β are roots quadratic equation

$$375x^2 - 25x - 2 = 0$$

The sum of roots is $\alpha + \beta = \frac{25}{375} = \frac{1}{15}$

The product of roots is $\alpha\beta = \frac{-2}{375}$

And, α & $\beta \in (-1, 1)$, then,

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n (\alpha^r + \beta^r)$$

$$\Rightarrow (\alpha + \alpha^2 + \dots \text{upto infinite term}) + (\beta + \beta^2 + \dots \text{upto infinite term})$$

$$\Rightarrow \frac{\alpha}{1-\alpha} + \frac{\beta}{1-\beta} \quad \left[\because S_{\infty} = \frac{\alpha}{1-r} \text{ for GP} \right]$$

$$\Rightarrow \frac{\alpha(1-\beta) + \beta(1-\alpha)}{(1-\alpha)(1-\beta)} = \frac{(\alpha + \beta) - 2\alpha\beta}{1 - (\alpha + \beta) + \alpha\beta}$$

$$\Rightarrow \frac{\frac{1}{15} + \frac{4}{375}}{1 - \frac{1}{15} - \frac{2}{375}} = \frac{1}{12}$$

Thus, if α and β are the roots of the equation

$375x^2 - 25x - 2 = 0$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$ is equal to $\frac{1}{12}$

67. The value of $\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$ is equal to:

A. $\pi - \sin^{-1}\left(\frac{63}{65}\right)$

B. $\frac{\pi}{2} - \cos^{-1}\left(\frac{9}{65}\right)$

C. $\frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$

D. $\pi - \cos^{-1}\left(\frac{33}{65}\right)$

Ans: C

It is given that

$$\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$$

Now,

$$\begin{aligned}
 & \sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right) \\
 &= \sin^{-1}\left[\frac{12}{13}\sqrt{1-\frac{9}{25}} - \frac{3}{5}\sqrt{1-\frac{144}{169}}\right] \\
 & \quad \left[\because \sin^{-1}x - \sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2} - y\sqrt{1-x^2}\right)\right] \\
 &= \sin^{-1}\left(\frac{33}{65}\right) \\
 &= \cos^{-1}\left(\frac{56}{65}\right) \\
 &= \frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)
 \end{aligned}$$

Thus,

The value of $\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$ is equal to $\frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$.

68. If the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the plane $2x + 3y - z + 13 = 0$ at a point P and the plane $3x + y + 4z = 16$ at a point Q, then PQ is equal to:
- A. $2\sqrt{14}$ B. $\sqrt{14}$
 C. 14 D. $2\sqrt{7}$

Ans: A

Equation of given line is

$$\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1} = \lambda$$

Now, coordinates of a general point line is

$$(3\lambda + 2, 2\lambda - 1, -\lambda + 1)$$

Since, p is the point of intersection and the plane

$$2x + 3y - z + 13 = 0 \text{ then}$$

$$2(3\lambda + 2) + 3(2\lambda - 1) - (-\lambda + 1) + 13 = 0$$

$$\lambda = -1$$

So, point P(-1, -3, 2)

Line intersect plane : $3x + y + 4z = 16$ at Q then

$$3(3\lambda + 2) + 2\lambda - 1 + 4(-\lambda + 1) = 16 \Rightarrow \lambda = 1$$

So, point Q(5, 1, 0) then $PQ = \sqrt{(5+1)^2 + (1+3)^2 + (0-2)^2}$

$$PQ = 2\sqrt{14}$$

69. If A is a symmetric matrix and B is a skew-symmetric matrix such that

$A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to :

A. $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

B. $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

C. $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$

D. $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

Ans: B

Given, matrix A is symmetric and matrix B is a skew-symmetric

A is symmetric $A^T = A$

B is skew symmetry $B^T = -B$

Since, $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$... (i)

On taking transpose both sides, we get

$$(A + B)^T = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}^T$$

Transpose

$$A^T + B^T = \begin{bmatrix} 2 & 5 \\ 3 & -1 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 2 & 5 \\ 3 & -1 \end{bmatrix} \quad \dots (ii)$$

From (i) + (ii)

$$A = \begin{bmatrix} 2 & 4 \\ 4 & -1 \end{bmatrix}$$

From (i) - (ii)

$$B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{So, } AB = \begin{bmatrix} 2 & 4 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$$

70. The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is:

A. 4

B. 3

C. 5

D. 7

Ans: C

The Given equation is

$$1 + \sin^4 x = \cos^2 3x; x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$$

$$\Rightarrow \text{L.H.S.} \geq 1$$

$$\text{R.H.S.} \leq 1$$

Both satisfy when

$$\text{L.H.S.} = \text{R.H.S.} = 1$$

$$\text{Since, } x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$$

$$\sin^4 x = 0; \cos^2 3x = 1$$

Hence, the value of x is

$$x = -2\pi, -\pi, 0, \pi, 2\pi$$

Thus, there are five different values of x is possible.

71. The integral $\int \frac{2x^3 - 1}{x^4 + x} dx$ is equal to: (Here C is a constant of integration)

A. $\frac{1}{2} \log_e \frac{(x^3 + 1)^2}{|x^3|} + C$

B. $\log_e \frac{|x^3 + 1|}{x^2} + C$

C. $\frac{1}{2} \log_e \left| \frac{x^3 + 1}{x^2} \right| + C$

D. $\log_e \left| \frac{x^3 + 1}{x} \right| + C$

Ans: C

Let

$$I = \int \frac{2x^3 - 1}{x^4 + x} dx$$

$$\Rightarrow I = \int \frac{(4x^3 + 1) - (2x^3 + 2)}{x^4 + x} dx$$

$$\Rightarrow I = \int \frac{4x^3 + 1}{x^4 + x} dx - 2 \int \frac{1}{x} dx$$

Now, putting $x^4 + x = t$ and differentiating, we get,

$$\Rightarrow (4x^3 + 1)dx = dt$$

Now,

$$I = \int \frac{dt}{t} - 2 \int \frac{1}{x} dx$$

$$\Rightarrow I = \ln |t| - 2 \ln x + c$$

$$\Rightarrow I = \ln \left| \frac{x^4 + x}{x^2} \right| + C \Rightarrow I = \ln \left| \frac{x^3 + 1}{x} \right| + C$$

72. If $\int_0^{\frac{\pi}{2}} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx = m(\pi + n)$, then $m n$ is equal to :

A. 1

B. $\frac{1}{2}$

C. $-\frac{1}{2}$

D. -1

Ans: D

$$I = \int_0^{\pi/2} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx$$

$$I = \int_0^{\pi/2} \frac{\frac{\cos x}{\sin x}}{\frac{\cos x}{\sin x} + \frac{1}{\sin x}} dx$$

$$I = \int_0^{\pi/2} \frac{\cos x}{1 + \cos x} = \int_0^{\pi/2} \frac{2 \cos^2 \frac{x}{2} - 1}{2 \cos^2 \frac{x}{2}} dx$$

$$I = \int_0^{\pi/2} \left(1 - \frac{1}{2} \sec^2 x / 2 \right) dx$$

$$I = \left[x - \frac{2}{2} \tan(x / 2) \right]_0^{\pi/2}$$

$$\Rightarrow \left(\frac{\pi}{2} - 1 \right) = \frac{(\pi - 2)}{2} = \frac{1}{2} (\pi - 2)$$

Since, $I = m(\pi - n)$

on comparing both sides, we get

$$m = \frac{1}{2}, n = -2$$

Now, $m \cdot n = -1$

73. The equation $|z - i| = |z - 1|$, $i = \sqrt{-1}$, represents :

- A. A circle of radius $\frac{1}{2}$.
- B. The line through the origin with slope -1.
- C. A circle of radius 1.
- D. The line through the origin with slope 1.

Ans: D

$$|z - i| = |z - 1| \text{ (Given)}$$

$$\text{Let } z = x + iy$$

$$|x + i(y - 1)| = |(x - 1) + iy|$$

$$\Rightarrow x^2 + (y - 1)^2 = (x - 1)^2 + y^2$$

$$1 - 2x = 1 - 2y$$

$$\Rightarrow y = x$$

$$x - y = 0$$

This is equation of straight line with slope 1.

74. For $x \in \mathbb{R}$, let $[x]$ denote the greatest integer $\leq x$, then the sum of the series

$$\left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - \frac{1}{100}\right] + \left[-\frac{1}{3} - \frac{2}{100}\right] + \dots + \left[-\frac{1}{3} - \frac{99}{100}\right] \text{ is :}$$

A. -135

B. -153

C. -133

D. -131

Ans (C)

75. Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If value of y is 1 when $x = 1$. Then the value of x for which $y = 2$ is :

A. $\frac{3}{2} - \frac{1}{\sqrt{e}}$

B. $\frac{3}{2} - \sqrt{e}$

C. $\frac{5}{2} + \frac{1}{\sqrt{e}}$

D. $\frac{1}{2} + \frac{1}{\sqrt{e}}$

Ans B)

Explanation:

76. Let a random variable X have a binomial distribution with mean 8 and variance 4.

If $P(X \leq 2) = \frac{k}{2^n}$, then k is equal to :

A. 121

B. 17

C. 137

D. 1

Ans D)

Explanation:

77. If the data $x_1, x_2, \dots, x_{10}$ is such that the mean of first four of these is 11, the mean of the remaining six is 16 and the sum of squares of all of these is 2,000; then the standard deviation of this data is :

- A. $\sqrt{2}$ B. 2
C. 4 D. $2\sqrt{2}$

Ans B)

78. If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to :

- A. $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ B. $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$
C. $\left(\frac{1}{e}, \frac{1}{e^2}\right)$ D. $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$

Ans: D

$$e^y + xy = e \quad \dots(i)$$

Put $x = 0$ in (i)

$$\Rightarrow e^y = e \Rightarrow y = 1$$

Differentiate (i) wr. to x

$$e^y \frac{dy}{dx} + x \frac{dy}{dx} + y = 0 \quad \dots(ii)$$

Put $y = 1$ in (ii)

$$e \frac{dy}{dx} + 0 + 1 = 0 \Rightarrow \frac{dy}{dx} = -\frac{1}{e}$$

Differentiate (ii) w. r, to x

$$e^y \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot e^y \frac{dy}{dx} + x \frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{dy}{dx} = 0 \quad \dots(iii)$$

$$\begin{aligned} \text{Put } y = 1, x = 0, \frac{dy}{dx} &= -\frac{1}{e} \\ e \frac{d^2y}{dx^2} + \frac{1}{e} - \frac{2}{e} &= 0 \Rightarrow \frac{d^2y}{dx^2} = \frac{1}{e^2} \\ \Rightarrow \left(\frac{dy}{dx}, \frac{d^2y}{dx^2} \right) &= \left(-\frac{1}{e}, \frac{1}{e^2} \right) \end{aligned}$$

79. If m is the minimum value of k for which the function $f(x) = x\sqrt{kx - x^2}$ is increasing in the interval $[0, 3]$ and M is the maximum value of f in $[0, 3]$ when $k = m$, then the ordered pair (m, M) is equal to.

- A. $(3, 3\sqrt{3})$ B. $(4, 3\sqrt{3})$
C. $(4, 3\sqrt{2})$ D. $(5, 3\sqrt{6})$

Ans: B

$$\begin{aligned} f(x) &= x\sqrt{kx - x^2} = \sqrt{kx^3 - x^4} \\ f'(x) &= \frac{(3kx^2 - 4x^3)}{2\sqrt{kx^3 - x^4}} \geq 0 \text{ for } x \in [0, 3] \end{aligned}$$

$$\Rightarrow 3k - 4x \geq 0$$

$$3k \geq 4x$$

$$3k \geq 4x \text{ for } x \in [0, 3]$$

Hence, $k \geq 4$

i.e., $m = 4$

For $k = 4$,

$$\Rightarrow f(x) = x\sqrt{4x - x^2}$$

For max. value, $f'(x) = 0$

$$\Rightarrow x = 3$$

$$\text{i.e., } y = 3\sqrt{3}$$

$$\text{Hence, } M = 3\sqrt{3}$$

80. The equation $y = \sin x \sin(x + 2) - \sin^2(x + 1)$ represents a straight line laying in :

- A. First, second and fourth quadrants
- B. First, third and fourth quadrants
- C. Third and fourth quadrants only
- D. Second and third quadrants only

Ans: C

It is given that

$$y = \sin x \sin(x + 2) - \sin^2(x + 1)$$

Multiply and dividing the above equation by 2, we get,

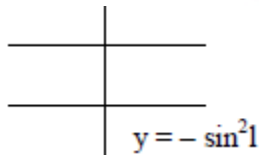
$$y = \frac{1}{2}[2 \sin x \sin(x + 2)] - \frac{1}{2}[2 \sin^2(x + 1)]$$

$$y = \frac{1}{2}[\cos(2) - \cos(2x + 2)] - \frac{1}{2}[1 - \cos 2(x + 1)]$$

$$y = \frac{1}{2}[\cos 2 - 1]$$

$$y = (-)\frac{1}{2}\sin^2 1$$

$$y = -\sin^2 1$$



Thus, the graph of y lies in IIIrd & IVth quadrant.

81. If the truth value of the statement $p \rightarrow (\sim q \vee r)$ is false (F), then the truth values of the statements p, q, r are respectively:

- A. T, F, T B. F, T, T
C. T, F, F D. T, T, F

Ans: D

It is given that

$p \rightarrow (\sim q \vee r)$ is false

It is true when

$p \rightarrow T \ \& \ (\sim q \vee r) = \text{false}$

It will true : $\sim q$ false & r false

$\sim q \rightarrow F \mid r \rightarrow F$

$\Rightarrow q \rightarrow T$

Truth value of p, q, r $\Rightarrow$ T, T, F

82. The number of ways of choosing 10 objects out of 31 objects of which 10 are identical and the remaining 21 are distinct, is:

- A. 2^{21} B. 2^{20}
C. $2^{20} + 1$ D. $2^{20} - 1$

Ans: B

Given, that, out of 31 objects 10 are identical and remaining 21 are distinct, so in following ways, we can choose 10 objects.

0 identical + 10 distincts, number of ways

$$= 1 \times {}^{21}C_{10}$$

1 identical + 9 distincts, number of ways

$$= 1 \times {}^{21}C_9$$

2 identical + 8 distincts, number of ways

$$= 1 \times {}^{21}C_8$$

10 identicals + 0 distinct, number of ways

$$= 1 \times {}^{21}C_0$$

So, total number of ways in which we can choose 10 objects is

$${}^{21}C_{10} + {}^{21}C_9 + {}^{21}C_8 + \dots + {}^{21}C_0 = x \text{ (let) } \dots (i)$$

$$\Rightarrow {}^{21}C_{11} + {}^{21}C_{12} + {}^{21}C_{13} + \dots + {}^{21}C_0 = x \dots (ii)$$

On adding both equation (i) and (ii) we get

$$2x = {}^{21}C_0 + {}^{21}C_1 + {}^{21}C_2 + \dots + {}^{21}C_{10} + {}^{21}C_{11} + {}^{21}C_{12} + \dots + {}^{21}C_{21}$$

$$\Rightarrow 2x = 2^{21}$$

$$\Rightarrow x = 2^{20}$$

83. For $x \in (0, 3/2)$ let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and $h(x) = \frac{1-x^2}{1+x^2}$. If

$\phi(x) = ((hof)og)(x)$, then $\phi\left(\frac{\pi}{3}\right)$ is equal to :

A. $\tan \frac{11\pi}{12}$

B. $\tan \frac{5\pi}{12}$

C. $\tan \frac{\pi}{12}$

D. $\tan \frac{7\pi}{12}$

Ans: A

Given that

For $x \in (0, 3/2)$, functions

$$f(x) = \sqrt{x}$$

$$g(x) = \tan x$$

$$\text{and } h(x) = \frac{1-x^2}{1+x^2}$$

Now,

$$\phi(x) = ((hof)og)(x) = (hof)(g(x))$$

$$= h(f(g(x))) = h(f(\tan x))$$

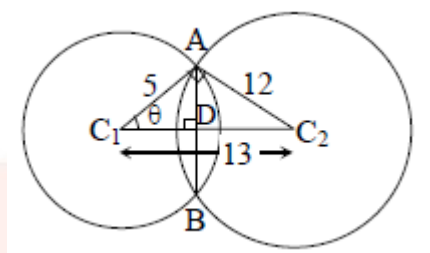
$$= h\left(\sqrt{\tan x}\right) = \frac{1 - \left(\sqrt{\tan x}\right)^2}{1 + \left(\sqrt{\tan x}\right)^2}$$

$$= \frac{1 - \tan x}{1 + \tan x}$$

C. $\frac{13}{2}$

D. $\frac{120}{13}$

Ans: D



In $\triangle AC_1C_2$

$$\tan \theta = \frac{12}{5} \Rightarrow \sin \theta = \frac{12}{13} \quad \dots(i)$$

In $\triangle ACD$:

$$\sin \theta = \frac{AD}{AC_1} \quad \dots(ii)$$

from (i) & (ii)

$$\Rightarrow \frac{AD}{5} = \frac{12}{13}$$

$$AD = \frac{120}{13}$$

Length of common chord (AB) = $\frac{120}{13}$

86. A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec., then the rate (in cm/sec) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is :

A. $\frac{25}{3}$

B. 25

C. $25\sqrt{3}$

D. $\frac{25}{\sqrt{3}}$

Ans: D

Given, $\frac{dy}{dt} = -25$ at $y = 1$

$$x^2 + y^2 = 4 \quad \dots(1)$$

On differentiating both sides of Equation (1) w.r.t. t.

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\frac{dx}{dt} = -\frac{y}{x} \frac{dy}{dt}$$

$$\text{At } y = 1; x = \sqrt{3}$$

$$(\because x^2 + y^2 = 4 \Rightarrow x^2 + 1 = 4 \Rightarrow x = \sqrt{3})$$

then

$$\frac{dx}{dt} = -\frac{1}{\sqrt{3}} \text{ cm / sec}$$

87. Let P be the point of intersection of the common tangents to the parabola $y^2 = 12x$ and the hyperbola $8x^2 - y^2 = 8$. If S and S' denote the foci of the hyperbola where S lies on the positive x – axis then P divides SS' in a ratio:

A. 14: 13

B. 5: 4

C. 2: 1

D. 13: 11

Ans: B

Equation of given parabola $y^2 = 12x \quad \dots(i)$

and hyperbola $8x^2 - y^2 = 8 \quad \dots(ii)$

Now, equation of tangent to parabola $y^2 = 12x$ having slope 'm' is $y = mx + 3/m \quad \dots(iii)$

and equation of tangent to hyperbola

$$\frac{x^2}{1} - \frac{y^2}{8} = 1 \text{ having slope 'm' is}$$

$$y = mx \pm \sqrt{1^2 m^2 - 8} \quad \dots(iv)$$

Since, tangents (iii) and (iv) represent the same line

$$\therefore m^2 - 8 = \left(\frac{3}{m}\right)^2$$

$$\Rightarrow m^4 - 8m^2 - 9 = 0$$

$$\Rightarrow (m^2 - 9)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 3$$

Now, equation of common tangents to the parabola (i) and hyperbola (ii) are

$$y = 3x + 1 \text{ and } y = -3x - 1$$

$\therefore$ Point 'P' is point of intersection of above common tangents,

$$\therefore P(-1/3, 0)$$

and focus of hyperbola S(3, 0) and S'(-3, 0).

Thus, the required ratio

$$= \frac{PS}{PS'} = \frac{3 + 1/3}{3 - 1/3} = \frac{10}{8} = \frac{5}{4}$$

88. Let S_n denote the sum of the first n terms of an A.P..

If $S_4 = 16$ and $S_6 = -48$, the S_{10} is equal to :

A. - 380

B. - 320

C. - 260

D. - 410

Ans: B

S_n = Sum of n terms of an A.P.

Given, $S_4 = 16 = a + 3d$ (i)

and $S_6 = -48 = a + 5d$ (ii)

On solving equation (i) & (ii), we get,

$$d = -32 \text{ } a = 112$$

$$S_{10} = \frac{10}{2} [2.(112) + (10 - 1)(-32)] = 5[-64]$$

$$S_{10} = -320$$

Thus, the S_{10} is equal to -320.

89. The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is:

- A. 126 B. -126
C. 84 D. -84

Ans: C

Given expression is

$$\begin{aligned} & (1+x)(1-x)^{10}(1+x+x^2)^9 \\ &= (1+x)(1-x)[(1-x)(1+x+x^2)]^9 \\ &= (1-x^2)(1-x^3)^9 \end{aligned}$$

Now, coefficient of x^{18} in the product

$$\begin{aligned} & (1+x)(1-x)^{10}(1+x+x^2)^9 \\ &= \text{coefficient of } x^{18} \text{ in the product} \\ & (1-x^2)(1-x^3)^9 \\ &= \text{coefficient of } x^{18} \text{ in } (1-x^3)^9 - \text{coefficient of } x^{16} \text{ in } (1-x^3)^9 \end{aligned}$$

Since, $(r+1)$ th term in the expansion of

$$(1-x^3)^9 \text{ is } {}^9C_r(-x^3)^r = {}^9C_r(-1)^r x^{3r}$$

Now, for x^{18} , $3r = 18 \Rightarrow r = 6$

and for x^{16} , $3r = 16$

$$\Rightarrow r = \frac{16}{3} \notin \mathbb{N}.$$

$$\therefore \text{required coefficient is } {}^9C_6 = \frac{9!}{3!3!} = \frac{9 \times 8 \times 7}{3 \times 2} = 84$$

90. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function such that $f(2) = 6$ and $f'(2) = \frac{1}{48}$. IF $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$, then

$\lim_{x \rightarrow 2} g(x)$ is equal to:

- A. 18 B. 36
C. 12 D. 24

Ans: A

Given that

$$\int_6^{f(x)} 4t^3 \cdot dt = (x-2)g(x); f(2) = 6; f'(2) = \frac{1}{48}$$

$$g(x) = \frac{\int_6^{f(x)} 4t^3 dt}{(x-2)}$$

$$\lim_{x \rightarrow 2} \frac{\int_6^{f(x)} 4t^3 dt}{(x-2)}$$

$$\lim_{x \rightarrow 2} \frac{4[f(x)]^3 \cdot f'(x)}{1}$$

At $x = 2$,

$$\lim_{x \rightarrow 2} g(x) = 4[f(2)]^3 \cdot f'(2)$$

$$\Rightarrow 4(6)^3 \left(\frac{1}{48} \right) = 18$$